A Smaller Obstruction to Polynomial Ordered Binary Matrix Removal

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Candidate research manuscript. Complete written argument and exact finite checks; independent mathematical review, end-to-end formalization, and novelty assessment remain pending. Developed with Codex assistance.

Abstract

We propose an explicit fixed 15-by-15 binary pattern with no polynomial ordered induced removal bound. The construction adapts OpenAI's 66-by-66 obstruction by proving a sharper three-column nonshattering lemma for its tree host and replacing the 64-by-64 anchor with a 13-by-13 anchor. The hosts preserve independently increasing row and column orders, require both zero and one constraints, and permit arbitrary binary changes in either direction. For every integer $h \geq 1$ the host has order $(80h+2)2^h$, normalized repair distance at least $(80h+2)^{-2}$, and exactly $2 \cdot 2^{28h}$ ordered copies. These give the claimed asymptotic failure for every fixed polynomial. The finite checks do not themselves prove the universal claim.

Source and contribution

The tree host, four-step propagation mechanism, sampling argument against arbitrary repairs, and diagonal-leaf localization method are adapted from OpenAI, *Polynomial removal fails for ordered binary matrices*, September 25, 2026, at commit `fd4aeeb2ee4fc729c18d98444fed42fd0529eeeb`. The proposed new ingredients are the three-column nonshattering lemma and the 13-by-13 anchor rigidity argument. This is a smaller-pattern refinement, not a new proof of the already claimed qualitative failure of polynomial removal. The exact 4^-h decay can also be derived for the original host. See the baseline audit and attribution. The original 66-by-66 Lean implementation has not been rebuilt here, and its formalization does not certify the new 15-by-15 pattern.

Statement

There is an explicit fixed binary matrix H of order 15 such that, for every integer h >= 1, with m = 2^h, d = 80h+2 and n = dm, there is a binary n-by-n matrix A_h satisfying

$$\operatorname{dist}_H(A_h) \geq d^{-2}, \qquad N_H(A_h) = 2m^{28}, \qquad \frac{N_H(A_h)}{n^{30}} = 2d^{-30}4^{-h}.$$

Consequently, for all real c,C > 0 there exist n, 0 < epsilon < 1 and A with dist_H(A) >= epsilon and N_H(A) < c epsilon^C n^30. The pattern is fixed before c,C,h are chosen. The density normalization is n^30, not the number of possible submatrices; this is the baseline convention.

1. Explicit pattern

Indices in this proof are one based, except for tree nodes and word labels. Let s=13. For 1 <= u,v <= 13 put

$$S(u,v) = \begin{cases} \lfloor (u-6)/2^{13-v} \rfloor \bmod 2, & 6 \leq u \leq 13, \quad 11 \leq v \leq 13, \\ \mathbf{1}[u \neq v], & \text{otherwise.} \end{cases}$$

Thus rows 6 through 13 on columns 11 through 13 list 000,001,...,111. With e_1,e_2 the first two standard unit vectors in {0,1}^13, put

$$H = \begin{pmatrix} S & e_1 & e_2 \\ e_1^T & 1 & 0 \\ e_2^T & 1 & 1 \end{pmatrix}, \qquad P = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}.$$

S has the following checked elementary properties:

1. Its rows are distinct: rows 1 through 10 have distinct zero positions in columns 1 through 10; rows 11 through 13 are all one there and have distinct last-three-bit words 101,110,111.
2. Its columns are distinct: columns 1 through 10 have single distinct zeros. Each of the last three columns has four zeros, and their word-coordinate vectors are distinct.
3. Every column has at most one zero in rows 1 through 6. For columns 1 through 10 this is the diagonal zero, if present. For columns 11 through 13 it is precisely the zero in row 6 (word 000).
4. Every row has at most one zero in columns 1 through 6, since this region avoids the replacement columns.
5. Every row and column has at least nine ones. Rows 1 through 10 have at least nine among their first ten entries; the other rows have ten there. The last-three columns have nine ones in total.

In particular every column has at least five ones in the first six rows, and every row has at least five in the first six columns. Any two columns share a one in the first six rows, and any two rows share a one in the first six columns (at most two excluded positions).

2. Host definition

At depth i of a full binary tree label nodes p=0,...,2^i-1. At every depth i<h introduce row blocks R_i^-(p), R_i^+(p) and column blocks C_i^-(p), C_i^+(p), each of size m/2^i. At depth h there is a single row and column position per node, given both sign names:

$$R_h^-(p) = R_h^+(p), \qquad C_h^-(p) = C_h^+(p).$$

All other blocks on an axis are disjoint. R_i^sigma and C_i^sigma denote unions over nodes at a depth; each has size m. For i>=1 the subscript (i,d) restricts to nodes with p congruent to d modulo 2. There are also dummy blocks R_, $C_$ of size m.

Order variable rows recursively: a node's minus block, its child-0 subtree, its child-1 subtree, then its plus block. At a leaf insert the one block once. Order columns similarly, with plus before and minus after the subtrees. Put the dummy rows after variable rows, and dummy columns before variable columns.

For every i=1,...,h introduce these six modes:

Mode	Row role 1	Row role 2	Column role 1	Column role 2
V_i^+	R_i^+	R_{i-1}^+	C_*	C_{i-1}^+
V_i^-	R_{i-1}^-	R_i^-	C_*	C_{i-1}^-
W_{i,0}^+ and W_{i,1}^+	R_{i,d}^+	R_*	C_{i-1}^+	C_{i,d}^+
W_{i,0}^- and W_{i,1}^-	R_{i,d}^-	R_*	C_{i,d}^-	C_{i-1}^-

Place one anchor block per mode before every non-anchor position on each axis. Within an anchor block put s consecutive groups of size m. Use increasing i, then V^+,V^-,W_0^+,W_1^+,W_0^-,W_1^- to order modes. Each mode's two roles on a given axis are disjoint, even at leaves.

Define every host entry as follows:

- Between anchor groups in the same mode put the corresponding S entry; between different modes put zero.
- A non-anchor position in role j of mode t has the unit-vector signature e_j against that mode's anchor groups. A position in neither role has the zero signature. Apply this independently on both axes.
- A non-anchor-to-non-anchor entry touching a dummy position is one.
- Variable-to-variable entries are zero except for these rules:

Row	Column	Range	Entry
R_i^+(p)	C_i^+(q)	0<=i<=h	1[p<=q]
R_i^-(p)	C_i^-(q)	0<=i<h	1[p<q]
R_i^+(p)	C_{i-1}^+(q)	1<=i<=h	1[floor(p/2)<=q]
R_i^-(p)	C_{i-1}^-(q)	1<=i<=h	1[floor(p/2)<q]

The common leaves satisfy the first rule. The second excludes them; in the final two rules the column signs are still distinct. Hence these assignments are consistent. There are 6hs anchor classes, 2h+1 variable classes and one dummy class of size m on each axis, giving n=(6hs+2h+2)m=(80h+2)m. No labels are part of the binary alphabet: the block names serve only to describe actual positions and entries.

The recursive orders imply, writing pi(p)=floor(p/2),

$$R_i^+(p) < R_{i-1}^+(a) \iff \pi(p) \le a, \quad R_{i-1}^-(a) < R_i^-(p) \iff a \le \pi(p),$$
$$C_{i-1}^+(b) < C_i^+(q) \iff b \le \pi(q), \quad C_i^-(q) < C_{i-1}^-(b) \iff \pi(q) \le b.$$

This follows because the subtrees at a fixed depth occur as consecutive intervals in increasing node order, and each child's block lies between its own parent's two signs. These comparisons include the shared leaves.

3. Stronger non-anchor trace lemma

Lemma. No three non-anchor columns are shattered by the non-anchor rows, for any h>=1. Columns need not be in increasing order.

Proof. A dummy column is constantly one, so cannot belong to a shattered set. Otherwise classify columns by level and sign, counting the common leaf class L only once. In a class the ones of every variable row form a suffix when columns are sorted by node; dummy rows are all one. Thus two columns from the same class cannot be shattered: the word one on the earlier and zero on the later cannot occur. Equal node indices or repeated positions also cannot be separated.

It remains to treat three distinct classes. Define a graph on the column classes: join (i,sigma) to (i-1,sigma) for 1<=i<h, and make a triangle on L,(h-1,+),(h-1,-). These are two paths ending at that triangle. For a non-leaf variable row, the only possible nonzero column classes are its own class and the immediately preceding class of the same sign. For a leaf row they are exactly among

L,(h-1,+),(h-1,-). Consequently if a variable row is one in two distinct classes those classes are adjacent in this graph.

If three distinct classes were shattered, each two-one/one-zero word would require a variable row (a dummy row cannot supply the zero). Every pair of the three classes would therefore be adjacent. The only triangle is L,(h-1,+),(h-1,-): a path contains no triangle, and no earlier-level vertex has two neighbors in the terminal triangle.

Now select columns from those three classes, with node labels q,b_+,b_- respectively. Only leaf rows can be one in the leaf column. A leaf row with node p has the three entries

$$(\mathbf{1}[p \le q],\ \mathbf{1}[\lfloor p/2 \rfloor \le b_+],\ \mathbf{1}[\lfloor p/2 \rfloor < b_-]).$$

These traces are coordinatewise nonincreasing as p increases, so form a chain under coordinatewise comparison. They cannot include both (1,1,0) and (1,0,1), which are incomparable. Non-leaf rows have zero in the first coordinate, and dummy rows give (1,1,1). Thus the full trace family cannot contain both words either. This contradicts shattering and proves the lemma. QED.

The strengthening is sharp for this host: for h>=2, root-plus column node 0 and level-1 plus column node 0 are shattered. Rows at root plus and level-1 plus node 0 give 10 and 11. At h=2, leaf nodes 0 and 2 give 01 and 00; at h>2, level-2 plus nodes 0 and 2 give those words. Thus two-column nonshattering is false.

4. Rigidity of the smaller anchor

A variable row or column belongs to the roles of at most four modes. For a non-leaf row at depth i, these are V_iˆsigma,V_{i+1}ˆsigma and W_{i,d}ˆsigma, omitting out-of-range levels. A leaf row belongs to V_hˆ+,V_hˆ- and W_{h,d}ˆ+,W_{h,d}ˆ-. A non-leaf column belongs to V_{i+1}ˆsigma,W_{i+1,0}ˆsigma,W_{i+1,1}ˆsigma,W_{i,d}ˆsigma; a leaf column belongs only to W_{h,d}ˆ+,W_{h,d}ˆ-. Against a collection using at most one representative per anchor group, each variable position therefore has at most four ones.

Lemma. Every ordered copy of S in A_h uses exactly one representative of each anchor group of a single mode on each axis, in group order.

Proof. Two positions in the same anchor group are twins throughout the host. Dummy positions on the same axis are also twins. Since S has distinct rows and columns, a copy uses at most one position from each anchor group and at most one dummy per axis.

If a copy used at most five anchor rows and at most ten anchor columns, its last eight rows and last three columns would all be non-anchor: anchors precede all non-anchors. They would realize all eight words by S's word rectangle, contradicting Section 3. Thus either at least six selected rows or at least eleven selected columns are anchors.

First suppose at least six selected rows are anchors, so the first six are anchors. A selected variable column has at most four ones against these representatives, but every S column has at least five ones there. There are no variable columns. Since at most one selected column is dummy, at least twelve selected columns are anchors. Any two of them share a one in the first six selected rows, and hence belong to the same mode t (different modes have zero anchor incidences). An anchor row of another mode would have at most one one in the copy, contradicting the minimum nine ones of every S row. Thus all selected anchor rows belong to t.

A dummy column has at most one one against the first six selected anchor rows of this mode, since its signature is a unit vector or zero. This contradicts the required five ones. Therefore all columns are anchors of t. Any non-anchor row then has at most one one against them, so all rows are anchors of t too.

If instead there are at least eleven selected anchor columns, use the first six selected columns and interchange the axes in the argument: each S row has at least five ones there; variable rows have at most four; at least twelve rows are anchors; any two share a one among those six columns; the remaining dummy row and non-anchor columns are excluded by unit-vector sparsity and the minimum row/column density. Thus this case also forces both axes into a single mode.

There are thirteen selected positions and exactly thirteen anchor groups per axis in that mode, with no group repeated. Every group is therefore represented, in order. QED.

Corollary. In every H copy, the first thirteen rows and columns belong to one mode's anchor. The last two rows and columns belong to that mode's roles 1 and 2 respectively. Indeed their anchor signatures must be e_1,e_2. An anchor from another mode gives zero; an anchor from the same mode gives a dense S row/column; and only the designated non-anchor role j gives e_j. This proves the corollary for both axes.

5. Localization and exact number of copies

Consider an H copy and its mode, using the role corollary. Its body must be P. Write p for its child-level row node, q for its child-level column node, a for its parent-level row node and b for its parent-level column node, as applicable.

- In V_iˆ+, row order gives pi(p)<=a. The bottom-right one gives a<=b. The top-right entry is consequently one, contradicting P.
- In V_iˆ-, row order gives pi(p)>=a. The top-right zero gives a>=b. The bottom-right entry is consequently zero, contradicting P.
- In W_{i,d}ˆ+, column order gives b<=pi(q), and the top-left one gives pi(p)<=b. Since p,q have the same parity, p<=q. The top-right entry is one, contradicting P. This applies at the leaves too.
- In W_{i,d}ˆ-, column order gives pi(q)<=b, and the top-right zero gives pi(p)>=b. Common parity implies p>=q. For i<h the top-left entry 1[p<q] is zero. For i=h it is instead 1[p<=q], and can be one only if p=q. The parent column then has b=pi(p).

Conversely, in W_{h,d}ˆ- each leaf p of parity d, any dummy row, and any parent-minus column in C_{h-1}ˆ-(pi(p)) produce exactly P, in the required order. Any representatives of that mode's thirteen anchor groups complete an H copy.

There are m leaf choices, m dummy row choices, two parent column choices, and mˆ{26} anchor choices. Each embedding has a unique mode and leaf, so

$$N_H(A_h) = m \cdot m \cdot 2 \cdot m^{26} = 2m^{28}.$$

Every copy sends the fixed pattern cell (14,14) into the m-cell leaf diagonal L_h. This also gives the coarser bound N_H(A_h)<=m nˆ{28}, which already suffices for nonpolynomial removal.

6. Distance against arbitrary binary repairs

Let B differ from A_h on a set E of M<mˆ2 cells. Partition each axis into the size-m anchor groups, dummy group, the two sign classes at each depth below h, and the single shared leaf class.

Choose a uniform leaf z in {0,...,m-1} and its ancestor a_i=floor(z/2ˆ{h-i}). Choose one uniform position x_iˆsigma,y_iˆsigma from the row and column blocks with that ancestor at each level and sign. At the leaf use one common position for both signs on each axis. Conditional on z, make all other choices independently. Also choose independently one uniform representative of every anchor and dummy group on each axis.

Each variable position has marginal sampling probability 2^{-i}/(m/2^i)=1/m. Anchor and dummy selections are independent of every opposite-axis selection. The root selections, whose node is always 0, are independent uniform choices too. Therefore every cell with an anchor or dummy endpoint, and every root-to-root cell, is selected with probability exactly 1/m^2. Within each class there is only one selection. A union bound over edited cells of these types gives probability at most M/m^2<1 that any such selected cell was edited. Fix a selection with none edited.

This statement does NOT apply to arbitrary variable-to-variable cells: for example a fixed diagonal leaf cell is selected with probability 1/m, because the two axes share z. No such independence is used below.

Suppose B were H-free. Its selected anchors still form S, and their signatures against selected body positions and all dummy incidences are unchanged. Thus whenever a mode's selected body is P, it extends to H in B. Its absence forces the following implications.

The unchanged root-plus diagonal is one and root-minus diagonal is zero. Suppose at depth i-1 these values are still one and zero. The V_i^+ and V_i^- bodies respectively have form

$$\begin{pmatrix} 1 & b_+ \\ 1 & 1 \end{pmatrix}, \qquad \begin{pmatrix} 1 & 0 \\ 1 & b_- \end{pmatrix}.$$

Avoiding P forces b_+=1 and b_-=0. These are the entries from selected depth-i rows to selected depth-(i-1) columns. Next use the W modes with d=a_i mod 2. Their bodies respectively have form

$$\begin{pmatrix} 1 & c_+ \\ 1 & 1 \end{pmatrix}, \qquad \begin{pmatrix} c_- & 0 \\ 1 & 1 \end{pmatrix}.$$

Avoiding P forces c_+=1 and c_-=0 at the depth-i diagonals. All four bodies are in increasing row/column order because their selected nodes are parent and child, and dummy positions have the stipulated extreme orders. Induction carries opposite values to depth h. At that depth the plus and minus diagonals are the same cell, a contradiction. Hence B is not H-free.

Every H-free repair requires at least m^2 changes. Dividing by n^2=d^2m^2 gives dist_H(A_h)>=d^{-2}. No restrictions were imposed on unprotected entries or on the directions of the changes.

7. Final asymptotic implication and sampling scope

For epsilon_h=d^{-2} the exact density gives, for every fixed C>0,

$$\frac{N_H(A_h)}{\epsilon_h^C n^{30}} = 2(80h+2)^{2C-30}4^{-h} \longrightarrow 0.$$

An exponential in h beats every fixed power of h; choosing h sufficiently large for any c>0 proves the theorem. Also 0<epsilon_h<1 for h>=1. The constants are uniform over the unbounded sequence, not just finitely many tested hosts.

For canonical sampling of q uniform rows and, independently, q uniform columns, rejection on observing H implies sampling one of L_h's m cells. Thus Pr(reject)<=m(q/n)^2=epsilon_h 2^{-h}q^2. Rejection probability at least rho>0 requires q>=sqrt(rho)(80h+2)2^{h/2}. Equivalently q>=sqrt(rho) epsilon_h^{-1/2} 2^{(epsilon_h^{-1/2}-2)/160} along this sequence. There are q^2 observed entries. This does not claim a lower bound for arbitrary adaptive or two-sided testers.

Changing every original leaf-diagonal one to zero destroys the old copies, but creates a new W_{h,d}^+ copy at each leaf: its parent-plus entry is one, the changed leaf entry is zero, and its

dummy bottom row is 11. This makes explicit why the m-cell hitting set is not an m-cell repair, and why both directions of binary edits must be allowed.

Verification boundary

The proof is all-depth mathematics; finite exact checks exercise its inputs and boundaries but are not its justification. Source hashes, anchor properties, non-anchor triples, mode bodies and weighted counts are checked by check_compression.py. No independent reviewer, Lean kernel build, or equivalence check of a new formal statement has been performed. The released 66-by-66 Lean proof is not a formal proof of this 15-by-15 construction.